

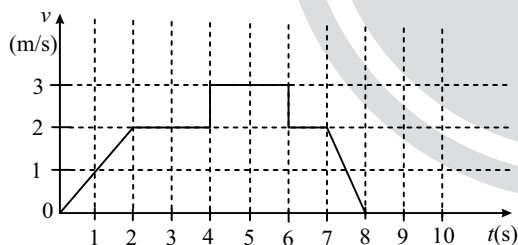
Motion in a Straight Line

Single Correct Type Questions

1. A particle moves from the point $(2.0\hat{i} + 4.0\hat{j})\text{m}$, at $t = 0$ with an initial velocity $(5.0\hat{i} + 4.0\hat{j})\text{ms}^{-1}$. It is acted upon by a constant force which produces a constant acceleration $(4.0\hat{i} + 4.0\hat{j})\text{ms}^{-2}$. What is the distance of the particle from the origin at time 2s? [11 Jan, 2019 (Shift-II)]

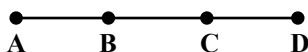
- (a) 15 m (b) $20\sqrt{2}$ m
(c) 5 m (d) $10\sqrt{2}$ m

2. A particle starts from the origin at time $t = 0$ and moves along the positive x -axis. The graph of velocity with respect to time is shown in figure. What is the position of the particle at time $t = 5\text{s}$? [10 Jan, 2019 (Shift-II)]



- (a) 10 m (b) 6 m (c) 3 m (d) 9 m

3. An object moves with speed v_1 , v_2 , and v_3 along a line segment AB, BC and CD respectively as shown in figure. Where $AB = BC$ and $AD = 3 AB$, then average speed of the object will be: [1 Feb, 2023 (Shift-I)]



- (a) $\frac{v_1 + v_2 + v_3}{3}$ (b) $\frac{v_1 v_2 v_3}{3(v_1 v_2 + v_2 v_3 + v_3 v_1)}$
(c) $\frac{3v_1 v_2 v_3}{v_1 v_2 + v_2 v_3 + v_3 v_1}$ (d) $\frac{(v_1 + v_2 + v_3)}{3v_1 v_2 v_3}$

4. A car travels a distance of 'x' with speed v_1 and then same distance 'x' with speed v_2 in the same direction. The average speed of the car is: [25 Jan, 2023 (Shift-I)]

- (a) $\frac{v_1 v_2}{2(v_1 + v_2)}$
(b) $\frac{v_1 + v_2}{2}$
(c) $\frac{2x}{v_1 + v_2}$
(d) $\frac{2v_1 v_2}{v_1 + v_2}$

5. The position of a particle related to time is given by $x = (5t^2 - 4t + 5)\text{m}$. The magnitude of velocity of the particle at $t = 2\text{s}$ will be: [15 April, 2023 (Shift-I)]

- (a) 10ms^{-1} (b) 14ms^{-1}
(c) 16ms^{-1} (d) 06ms^{-1}

6. The velocity of a particle is $v = v_0 + gt + Ft^2$. Its position is $x = 0$ at $t = 0$; then its displacement after time ($t = 1$) is: [17 March, 2021 (Shift-II)]

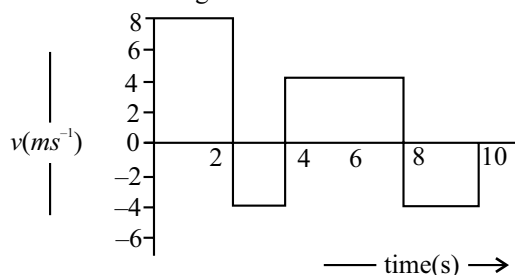
- (a) $v_0 + g + F$
(b) $v_0 + \frac{g}{2} + \frac{F}{3}$
(c) $v_0 + 2g + 3F$

- (d) $v_0 + \frac{g}{2} + F$

7. The position of a particle as a function of time ts , is given by $x(t) = at + bt^2 - ct^3$ where a , b and c are constants. When the particle attains zero acceleration, then its velocity will be: [9 April, 2019 (Shift-II)]

- (a) $a + \frac{b^2}{4c}$ (b) $a + \frac{b^2}{c}$
(c) $a + \frac{b^2}{2c}$ (d) $a + \frac{b^2}{3c}$

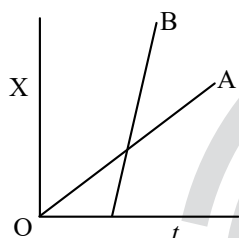
8. The velocity time graph of a body moving in a straight line is shown in figure.



The ratio of displacement to distance travelled by the body in time 0 to 10 s is [24 Jan, 2023 (Shift-II)]

- (a) 1 : 1 (b) 1 : 4 (c) 1 : 2 (d) 1 : 3

9. The position-time graphs for two students A and B returning from the school to their homes are shown in figure:



- (A) A lives closer to the school
(B) B lives closer to the school
(C) A takes lesser time to reach home
(D) A travels faster than B
(E) B travels faster than A [10 April, 2023 (Shift-I)]

Choose the correct answer from the options given below:

- (a) (A) and (E) only (b) (B) and (E) only
(c) (A), (C) and (E) only (d) (A), (C) and (D) only

10. An engine of a train, moving with uniform acceleration, passes the signal - post with velocity u and the last compartment with velocity v . The velocity with which middle point of the train passes the signal post is :

[25 Feb, 2021 (Shift-I)]

- (a) $\frac{u+v}{2}$
(b) $\sqrt{\frac{v^2+u^2}{2}}$
(c) $\frac{v-u}{2}$
(d) $\sqrt{\frac{v^2-u^2}{2}}$

11. In a car race on straight road, car A takes a time ' t ' less than car B at the finish and passes finishing point with a speed ' V ' more than that of car B. Both the cars start from rest and travel with constant acceleration a_1 and a_2 respectively. Then ' v ' is equal to [9 Jan, 2019 (Shift-II)]

- (a) $\frac{2a_1a_2}{a_1+a_2}t$ (b) $\sqrt{2a_1a_2}t$
(c) $\sqrt{a_1a_2}t$ (d) $\frac{a_1+a_2}{2}t$

12. A ball is thrown vertically upward with an initial velocity of 150 m/s. The ratio of velocity after 3s and 5s is $\frac{x+1}{x}$.

The value of x is [12 April, 2023 (Shift-I)]

Take ($g = 10 \text{ m/s}^2$).

- (a) 6 (b) 5
(c) -5 (d) 10

13. A juggler throws ball vertically upwards with same initial velocity in air. When the first ball reaches its highest positions, he throws the next ball. Assuming the juggler throws n balls per second, the maximum height the balls can reach is [29 July, 2022 (Shift-II)]

- (a) $g/2n$ (b) $2gn$
(c) g/n (d) $g/2n^2$

14. An object of mass 5 kg is thrown vertically upwards from the ground. The air resistance produces a constant retarding force of 10 N throughout the motion. The ratio of time of ascent to the time of descent will be equal to: [Use $g = 10 \text{ ms}^{-2}$]. [24 June, 2022 (Shift-II)]

- (a) 1 : 1 (b) $\sqrt{2} : \sqrt{3}$
(c) $\sqrt{3} : \sqrt{2}$ (d) 2 : 3

15. A balloon was moving upwards with a uniform velocity of 10 m/s. An object of finite mass is dropped from the balloon when it was at a height of 75m from the ground level. The height of the balloon from the ground when object strikes the ground was around: (takes the value of g as 10 m/s^2) [25 July, 2021 (Shift-II)]

- (a) 250 m (b) 300 m
(c) 200 m (d) 125 m

16. A stone is dropped from the top of a building. When it crosses a point 5m below the top, another stone starts to fall from a point 25m below the top. Both stones reach the bottom of building simultaneously. The height of the building is: [25 Feb, 2021 (Shift-II)]

- (a) 35 m (b) 45 m
(c) 50 m (d) 25 m

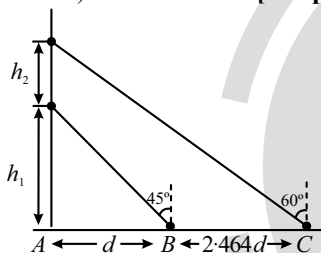
17. A ball is thrown up with a certain velocity so that it reaches a height 'h'. Find the ratio of the two different times of the ball reaching $h/3$ in both the directions:

[27 July, 2021 (Shift-I)]

- (a) $\frac{\sqrt{2}-1}{\sqrt{2}+1}$
 (b) $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$
 (c) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$
 (d) $\frac{1}{3}$

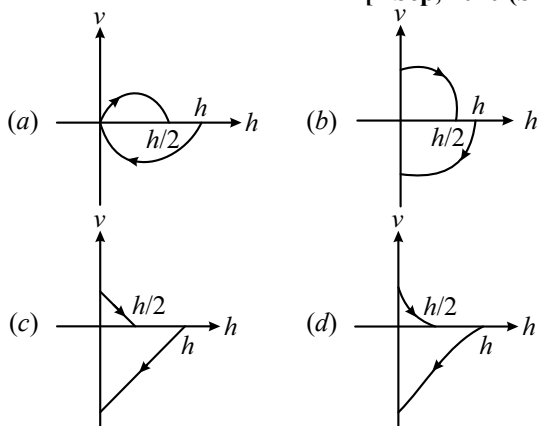
18. A balloon is moving up in air vertically above a point A on the ground. When it is at a height h_1 , a girl standing at a distance d (point B) from A (see figure) sees it at an angle 45° with respect to the vertical. When the balloon climbs up a further height h_2 , it is seen at an angle 60° with respect to the vertical if the girl moves further by a distance $2.464d$ (point C). Then the height h_2 is (given $\tan 30^\circ = 0.5774$)

[5 Sep, 2020 (Shift-I)]



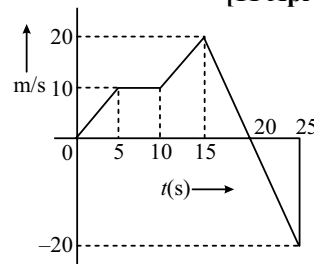
- (a) d (b) $0.732d$
 (c) $1.464d$ (d) $0.464d$
19. A tennis ball is released from a height h and after freely falling on a wooden floor it rebounds and reaches height $h/2$. The velocity versus height of the ball during its motion may be represented graphically by (graph are drawn schematically and on not to scale).

[4 Sep, 2020 (Shift-I)]



20. From the $v-t$ graph shown, the ratio of distance to displacement in 25 s of motion

[11 April, 2023 (Shift-I)]



- (a) $\frac{3}{5}$ (b) $\frac{1}{2}$
 (c) $\frac{5}{3}$ (d) 1

21. Match Column-I with Column-II:

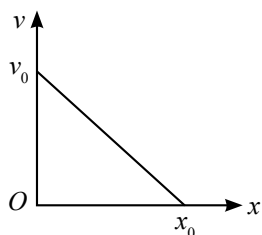
Column-I (x-t graphs)		Column-II (v-t graphs)	
A.		I.	
B.		II.	
C.		III.	
D.		IV.	

Choose the correct answer from the options given below:

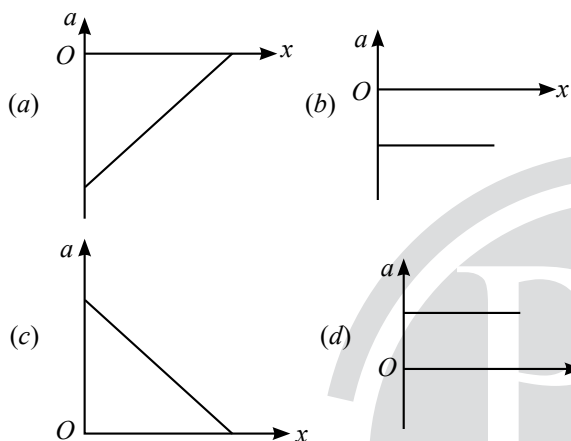
[30 Jan, 2023 (Shift-I)]

- (a) A- II, B-IV, C-III, D-I (b) A- I, B-II, C-III, D-IV
 (c) A- II, B-III, C-IV, D-I (d) A- I, B-III, C-IV, D-II

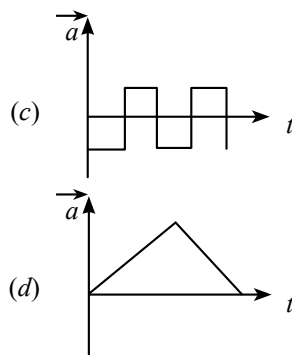
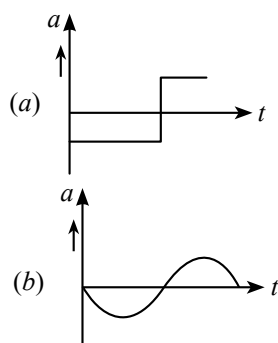
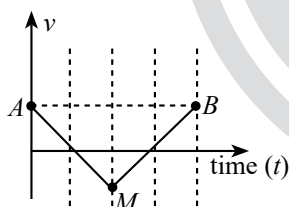
22. The velocity - displacement graph of a particle is shown in the figure. [18 March, 2021 (Shift-II)]



The acceleration - displacement graph of the same particle is represented by:



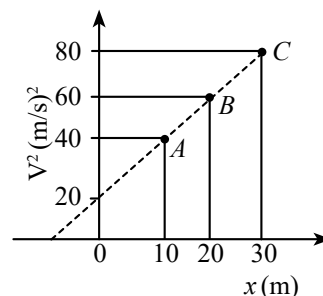
23. If the velocity-time graph has the shape AMB, what would be the shape of the corresponding acceleration-time graph? [24 Feb, 2021 (Shift-I)]



24. The relation between time t and distance x for a moving body is given as $t = mx^2 + nx$, where m and n are constants. The retardation of the motion is: (Where v stands for velocity) [25 July, 2021 (Shift-II)]
 (a) $2n^2v^2$ (b) $2mnv^3$ (c) $2mv^3$ (d) $2nv^3$

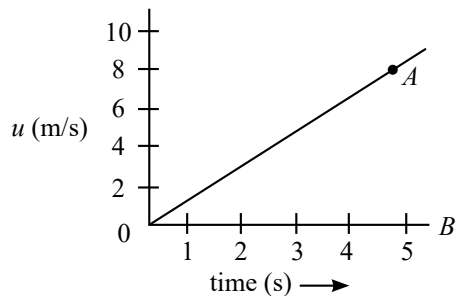
Integer Type Questions

25. A ball is thrown vertically upwards with a velocity of 19.6 ms^{-1} from the top of a tower. The ball strikes the ground after 6 s. The height from the ground up to which the ball can rise will be $\left(\frac{k}{5}\right) \text{ m}$. The value of k is _____. (use $g = 9.8 \text{ m/s}^2$) [28 July, 2022 (Shift-II)]
26. A tennis ball is dropped on to the floor from a height of 9.8 m . It rebounds to a height 5.0 m . Ball comes in contact with the floor for 0.2 s . The average acceleration during contact is _____ ms^{-2} [Given $g = 10 \text{ ms}^{-1}$] [29 Jan, 2023 (Shift-I)]
27. For a train engine moving with speed of 20 ms^{-1} , the driver must apply brakes at a distance of 500 m before the station for the train to come to rest at the station. If the brakes were applied at half of this distance, the train engine would cross the station with speed $\sqrt{x} \text{ ms}^{-1}$. The value of x is _____. (Assuming same retardation is produced by brakes) [1 Feb, 2023 (Shift-II)]
28. A particle is moving with constant acceleration ' a '. Following graph shows v^2 versus x (displacement) plot. The acceleration of the particle is _____ m/s^2 . [31 Aug, 2021 (Shift-II)]



29. The speed versus time graph for a particle is shown in the figure. The distance travelled (in m) by the particle during the time interval $t = 0$ to $t = 5$ s will be _____.

[3 Sep, 2020 (Shift-II)]



30. The distance x covered by a particle in one dimensional motion varies with time t as $x^2 = at^2 + bt + c$. If the acceleration of the particle depends on x as x^{-n} , where n is an integer, the value of n is _____.

[9 Jan, 2020 (Shift-I)]



ANSWER KEY

- | | | | | | | | | | |
|---------|---------|---------|---------|-----------|-----------|-----------|---------|----------|---------|
| 1. (b) | 2. (d) | 3. (c) | 4. (d) | 5. (c) | 6. (b) | 7. (d) | 8. (d) | 9. (a) | 10. (b) |
| 11. (c) | 12. (b) | 13. (d) | 14. (b) | 15. (d) | 16. (b) | 17. (b) | 18. (a) | 19. (b) | 20. (c) |
| 21. (a) | 22. (a) | 23. (a) | 24. (c) | 25. [392] | 26. [120] | 27. [200] | 28. [1] | 29. [20] | 30. [3] |

EXPLANATIONS

1. (b) $\vec{s} = (5\hat{i} + 4\hat{j})2 + \frac{1}{2}(4\hat{i} + 4\hat{j})4 = 10\hat{i} + 8\hat{j} + 8\hat{i} + 8\hat{j}$

$$\vec{r}_2 - \vec{r}_1 = 18\hat{i} + 16\hat{j}$$

$$\vec{r}_2 = 20\hat{i} + 20\hat{j}$$

$$|\vec{r}_f| = 20\sqrt{2}$$

2. (d) $r_{t=5} = \text{area}$

$$= \left(\frac{1}{2} \times 2 \times 2 + 2 \times 2 + 3 \times 1 \right) \text{m}$$

$$= (2 + 4 + 3) = 9 \text{ m.}$$

3. (c) Given $AB = x$ and $BC = x$

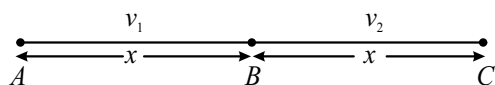
$$\therefore AB + BC + CD = 3x$$

$$\Rightarrow 2x + CD = 3x$$

$$\therefore CD = x$$

$$\text{Average speed} < v > = \frac{3x}{\frac{x}{v_1} + \frac{x}{v_2} + \frac{x}{v_3}} = \frac{3v_1v_2v_3}{v_2v_3 + v_1v_3 + v_1v_2}$$

4. (d)



$$\text{Average velocity} = \frac{\text{Total displacement of the car}}{\text{Total time}}$$

$$= \frac{x + x}{\frac{x}{v_1} + \frac{x}{v_2}}$$

$$= \frac{2v_1v_2}{v_1 + v_2}$$

5. (c) $x = 5t^2 - 4t + 5 \Rightarrow v = \frac{dx}{dt} = \frac{d}{dt}(5t^2 - 4t + 5)$

$$= 10t - 4$$

$$\text{At } t = 2\text{s and } v = 16\text{m/s}$$

6. (b) $V = V_0 + gt + Ft^2$

$$\frac{dx}{dt} = V_0 + gt + Ft^2 \Rightarrow \int_{x=0}^x dx = \int_{t=0}^{t=1} (V_0 + gt + Ft^2) dt$$

$$x = \left[V_0t + \frac{gt^2}{2} + \frac{Ft^3}{3} \right]_{t=0}^{t=1} \Rightarrow x = V_0 + \frac{g}{2} + \frac{F}{3}$$

7. (d) $x = at + bt^2 - ct^3$

$$v = \frac{dx}{dt} = a + 2bt - 3ct^2$$

$$a = \frac{dv}{dt} = 2b - 6ct = 0 \Rightarrow t = \frac{b}{3c}$$

$$v_{\left(t=\frac{b}{3c}\right)} = a + 2b\left(\frac{b}{3c}\right) - 3c\left(\frac{b}{3c}\right)^2$$

$$= a + \frac{b^2}{3c}$$

8. (d) Displacement of the body = $16 - 8 + 16 - 8 = 16\text{m}$

$$\text{Distance travelled by the body} = \Sigma |\text{area}| = 48 \text{ m}$$

$$\Rightarrow \frac{\text{displacement}}{\text{Distance}} = \frac{1}{3}$$

9. (a) As slope of B > Slope of A

$$\therefore \text{Speed of B} > \text{Speed of A}$$

$$\Rightarrow t_B < t_A$$

10. (b) Let the length of train be l and its acceleration be a .

From kinematic equation.

$$\Rightarrow al = \frac{v^2 - u^2}{2}$$

Velocity when middle point crosses the post,

$$V_m = \sqrt{u^2 + 2a \frac{l}{2}} = \sqrt{u^2 + \frac{v^2 - u^2}{2}} = \sqrt{\frac{u^2 + v^2}{2}}$$

$$11. (c) \sqrt{\frac{2\ell}{a_2}} - \sqrt{\frac{2\ell}{a_1}} = t \Rightarrow \frac{\sqrt{2\ell}}{t} = \frac{\sqrt{a_1 a_2}}{\sqrt{a_1} - \sqrt{a_2}}$$

$$\sqrt{2a_1 \ell} - \sqrt{2a_2 \ell} = v \Rightarrow \frac{\sqrt{2\ell}}{v} = \frac{1}{\sqrt{a_1} - \sqrt{a_2}}$$

$$\Rightarrow \frac{v}{t} = \sqrt{a_1 a_2} \Rightarrow v = (\sqrt{a_1 a_2})t$$

12. (b) Given, $u = 150 \text{ m/s}$

From kinematic equations, $\frac{1}{v} = \frac{1}{u} + \frac{1}{at}$

$$V = 150 - 10t$$

At, $t = 3$ seconds

$$V_1 = 150 - 30 = 120$$

At, $t = 5$ seconds

$$V_2 = 150 - 50 = 100$$

$$\frac{120}{100} = \frac{x+1}{x} \Rightarrow x = 5$$

13. (d) Time to reach at max height, $t = \frac{u}{g}$
No. of balls thrown in 1 sec = n .

So, time taken by each ball = $\frac{1}{n}$ sec

$$\text{i.e. } \frac{u}{g} = \frac{1}{n} \Rightarrow u = \frac{g}{n}$$

$$\text{So, } h_{\max} = \frac{u^2}{2g} = \frac{g^2}{2gn^2} = \frac{g}{2n^2}$$

14. (b) For time of ascent,

$$a = \frac{mg + 10}{g} = \frac{5 \times 10 + 10}{10} = 6 \text{ m/sec}^2$$

From formula, $v = u + at$

$$0 = u - 6t$$

$$t_A = \frac{u}{6} \quad \dots(i)$$

$$h = ut + \frac{1}{2}at^2$$

$$= u \times \frac{u}{6} - \frac{1}{2} \times 6 \times \frac{u}{6} \times \frac{u}{6}$$

$$= \frac{u^2}{6} - \frac{u^2}{12} = \frac{u^2}{12} \quad \dots(ii)$$

For time of descent,

$$a = \frac{50 - 10}{10} = 4 \text{ m/sec}^2$$

Using formula, $h = ut + \frac{1}{2}at^2$

$$\Rightarrow \frac{u^2}{12} = 0 \times t + \frac{1}{2} \times 4t^2 \quad [\text{From equation (ii)}]$$

$$\Rightarrow \frac{u^2}{24} = t^2$$

$$t_B = \frac{u}{\sqrt{24}} \quad \dots(iii)$$

On dividing equation (i) with equation (iii)

$$\frac{t_A}{t_B} = \frac{u \times \sqrt{24}}{6 \times u} = \frac{\sqrt{2}}{\sqrt{3}}$$

15. (d) Let time taken by the object to reach the ground be t .

Therefore, $s = ut + \frac{1}{2}at^2$

$$-75 = 10t - \frac{1}{2}gt^2$$

$$5t^2 - 10t - 75 = 0$$

$$t = 5 \text{ sec.}$$

Distance covered by balloon in 5 sec,

$$h = 10 \times 5 = 50 \text{ m}$$

$$\text{Total height of balloon} = 75 + 50 = 125 \text{ m}$$

16. (b) 5m below, velocity of first stone, $v = \sqrt{2gs} = 10 \text{ m/s}$

$$\therefore x + 20 = 10t + \frac{1}{2}gt^2$$

For second stone,

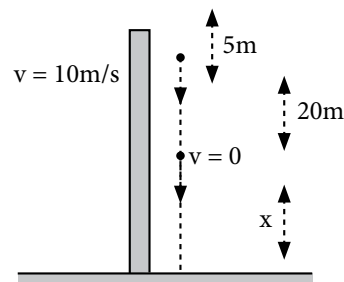
$$x = \frac{1}{2}gt^2$$

On solving these two equation, we get

$$t = 2 \text{ sec}$$

$$\therefore x = \frac{1}{2} \times 10 \times 4 = 20 \text{ m}$$

$$H = 20 + 5 = 45 \text{ m.}$$



17. (b) Speed of ball to reach a height h , $u = \sqrt{2gh}$

Given, $S = \frac{h}{3}$

Using kinematic equation, $S = ut + \frac{1}{2}at^2$

$$\frac{h}{3} = \sqrt{2gh}t - \frac{1}{2}gt^2$$

$$0 = t^2 \left(\frac{g}{2} \right) - \sqrt{2gh}t + \frac{h}{3}$$

$$t = \frac{\sqrt{2gh} \pm \sqrt{2gh - 4 \times \frac{g}{2} \times \frac{h}{3}}}{2 \times \frac{g}{2}}$$

$$\frac{t_1}{t_2} = \frac{\sqrt{2gh} - \sqrt{\frac{4gh}{3}}}{\sqrt{2gh} + \sqrt{\frac{4gh}{3}}}$$

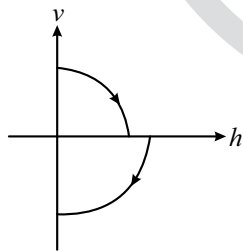
$$\frac{t_1}{t_2} = \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}}$$

18. (a) $\frac{h_1}{d} = \tan 45^\circ$

$$\frac{h_1 + h_2}{d + 2.464d} = \tan 30^\circ$$

$$\Rightarrow h_2 = d$$

19. (b) At $H = h$, $v = 0$



At $h = 0$, $v = \sqrt{2gh}$

Also $a = -g$, throughout this motion

20. (c) Area under the graph from $t = 0$ to $t = 20$ sec

$$A_1 = \frac{1}{2} \times 5 \times 10 + (5 \times 10) + \frac{1}{2} \times (10 + 20) \times 5 + \frac{1}{2} \times 5 \times 20 = 200 \text{ m}^2$$

Area under the graph from $t = 20$ to $t = 25$ sec

$$A_2 = \frac{1}{2} \times 5 \times (20) = 50 \text{ m}^2$$

So distance covered $= A_1 + A_2 = (200 + 50) \text{ m} = 250 \text{ m}$

Displacement $= A_1 - A_2 = (200 - 50) \text{ m} = 150 \text{ m}$

21. (a) $\frac{dx}{dt} = \text{slope} \geq 0$ always increasing

(A - II)

$$\frac{dx}{dt} < 0 ; \text{ and at } t \rightarrow \infty \frac{dx}{dt} \rightarrow 0$$

(B - IV)

$$\frac{dx}{dt} > 0 \text{ for first half } \frac{dx}{dt} < 0 \text{ for second half.}$$

(C - III)

$$\frac{dx}{dt} = \text{constant}$$

(D - I)

22. (a) The slope of the given v versus x graph is $m = -\frac{v_0}{x_0}$

and intercept is $c = +v_0$. Hence, varies with x as

$$v = -\left(\frac{v_0}{x_0}\right)x + v_0 \quad \dots(i)$$

Using equation (i) in equation (ii), we get

$$a = v \frac{dv}{dx} - \left(\frac{v_0}{x_0}\right) \left(-\frac{v_0}{x_0}x + v_0\right) \Rightarrow a = \left(\frac{v_0}{x_0}\right)^2 x - \frac{v_0^2}{x_0}$$

Thus the graph of a versus x is a straight line having

a positive slope $= \left(\frac{v_0}{x_0}\right)^2$ and negative intercept $= -\frac{v_0^2}{x_0}$.

23. (a) Slope of $v-t$ graph is constant but negative for part AM. Slope of $v-t$ graph is constant but positive for part MB. Hence, option (a) is the correct slope of the acceleration- time graph.

24. (c) Given, $t = mx^2 + nx$

$$\frac{dt}{dx} = 2mx + n \Rightarrow \frac{dx}{dt} = \frac{1}{2mx + n}$$

$$v = \frac{1}{2mx + n}$$

Retardation of the motion.

$$a = v \frac{dv}{dx} = v \frac{-2m}{(2mx + n)^2} = -2mv^3$$

25. [392] Applying, $s = ut + \frac{1}{2}at^2$

$$h = 19.6 \times 2 + \frac{1}{2} \times 9.8 \times 4 = 58.8 \text{ m}$$

Height above tower

$$h' = \frac{u^2}{2g} = \frac{19.6 \times 19.6}{2 \times 9.8} = 19.6$$

$$\therefore H_{(\text{from ground})} = 19.6 + 58.8 = 78.4\text{m}$$

$$\text{As, } \frac{k}{5} = 78.4 \Rightarrow k = 392$$

26. [120] Velocity of the ball just before reaching the ground,

$$v_i = \sqrt{2gh_i}$$

$$= 14\text{ m/s}$$

Velocity of the ball just after the rebound,

$$v_f = \sqrt{2gh_f}$$

$$= 10\text{ m/s}$$

$$\therefore \vec{a}_{\text{avg}} = \frac{\vec{v}_f - \vec{v}_i}{t}$$

$$\Rightarrow |\vec{a}_{\text{avg}}| = \left| \frac{\Delta \vec{v}}{\Delta t} \right| = 120\text{ m/s}^2$$

27. [200] $u = 20\text{ m/s}, S_1 = 500\text{ m}, v = 0$

Using third equation of motion

$$0 = (20)^2 - 2a \cdot 500$$

$$\Rightarrow a = \frac{4}{10}\text{ m/s}^2$$

After brakes are applied

$$u = 20\text{ m/s}, S_2 = 250\text{ m}, v = ?$$

$$v^2 = (20)^2 - 2a \cdot 250$$

$$= v = \sqrt{200}\text{ m/s}$$

$$\therefore x = 200$$

28. [1] As, $y = mx + c$

$$v^2 = 2x + 20$$

$$v^2 = 2x + 20$$

$$\Rightarrow 2v \frac{dv}{dt} = 2v$$

$$\Rightarrow a = 1\text{ m/s}^2$$

29. [20] Distance travelled = Area under the $u-t$ graph

$$\therefore \Delta S = \frac{1}{2} \times 5 \times 8 = 20$$

30. [3] We have given, the distance covered by particle varies with time t , $x^2 = at^2 + 2bt + c$

$$2xv = 2at + 2b$$

$$xv = at + b$$

$$v^2 + ax = a$$

$$ax = a - \left(\frac{at+b}{x} \right)^2$$

$$a = \frac{a(at^2 + 2bt + c) - (at+b)^2}{x^3}$$

$$a = \frac{ac - b^2}{x^3}; a \propto x^{-3}$$

Hence, value of n is 3.